FATE Foundation

UCL

AIMS African Institute for Mathematical Sciences SOUTH AFRICA

LECTURE IT BACK

Kenya - workshop

AI Research Foundations

with support from Google.org

Ten sessions

- Overview
- Check-in questionnaire
- Google DeepMind: AI Research Foundations
- Session 0
- Session 1
- Session 2
- Session 3
- Session 4
- Session 5
- Session 6
- Session 7
- Session 8
- Session 9
- Assessment
- How to use this toolkit
- Templates and design files
- License

How it works!

Plan

- Eight groups
- Each group will be assigned one AI activity or one RI Activity from the toolkit
- Each group prepares how you'll run them.

💡 **Remarks:**

- If the activities require handouts, they are provided with the activities.
- Need paper, markers, or anything else? **Just ask us** 🙌

Agenda :

Thursday 15 Oct (16:00 - 17 : 00)

Friday 16 Oct (16:00 - 17 : 00)

Structure of an activity

1 The header
Length, mode, type, Bloom level, key topics

Activity length	15 minutes
Mode	Group
Activity type (Conversational Framework)	Discussion
Bloom's Taxonomy Level	Understand
Linked online spotlight activity	What does it mean to train a model? Machine learning development pipeline
Key topics	Asymmetry; misclassification; loss; evaluation
Resources required	Discussion questions
Prepares students for	AI Medium Activity 1 - Loss matrix on a small dataset

2 Overview + objective
Why it exists, what it maps to online

Overview

This activity introduces students to the idea that evaluation is central to machine learning, and that some errors are more serious than others. This activity encourages reflection on why loss functions matter and how the choice of evaluation metric can entirely change the behavior of a model.

Learning objectives

- Maps to online course Google DeepMind: 01 Build Your Own Small Language Model learning objective(s):
    - o List the steps involved in the machine learning development pipeline, referencing key concepts such as tokenization, loss, learning, inference, and evaluation.
- Activity-specific learning objective(s):

Structure of an activity

3 The task
The script: what you say, what the room does

Task

1. Introduce the activity and show images of cheetahs, lions, and leopards on the screen.
   Introduce the concept of misclassification.

2. Put students into groups of 3-4.

3. Display the discussion questions and ask students to discuss them in their groups.

   Discussion questions:
   a. Why do we need evaluation in machine learning?

4 Teacher notes
Key points and facilitation guidance

Teacher notes

- Key points of learning:
  - Evaluation prevents models from being "blind" to their own errors.
  - Some errors have small consequences; others may be dangerous, costly, or ethically unacceptable. This might not be easy to spot if you are only focusing on a global error rate across all classes.
  - Rare classes can be overlooked if we only measure overall accuracy. This is known in the machine learning literature as the class imbalance problem.
  - Loss functions formalize the cost of making different mistakes.
  - Highlight that loss is not just a mathematical object: it expresses human priorities i.e. what errors we want to avoid. The choice of a loss function is therefore

In your delivery slot

Run

≈ Respect the timing

Deliver it as you would to your class: open the idea, run the task, manage the time, debrief the answers.

Feedback

≈ 5 minutes

Peers and facilitators respond: what landed, what to tighten (timing, clarity, participation, debrief).

Materials?

Deadline : Thursday - 10 a.m

Where? #lecture-it-back-kenya-cohort-1

AFRICA AI UPSKILLING PROGRAMME - AI RESEARCH FOUNDATIONS WORKSHOP

Lecture it Back · Group Assignments

Group 1 GROUP OF 4 — Thu 15 Oct · 16:00 to 16:20 · Room A

1.4 Loss matrix on a small dataset — AI · 20 MIN

Dr. Leah Mutanu
Dr. Henry Muchiri
Anthony Karugu
Bonsa Guyo

Group 2 GROUP OF 4 — Thu 15 Oct · 16:00 to 16:20 · Room B

2.4 Why tokenization matters — AI · 20 MIN

Prof. Jimmy Macharia
Dr. Mary Njeri Ngaruiya
Ernest Andugo
Serena Cheres

Group 3 TRIO — Fri 16 Oct · 16:00 to 16:20 · Room A

3.4 Where did the model go wrong? — AI · 20 MIN

Prof. Calvin Omolo
Dr. Vincent Mbandu
Michelle Chebet

Group 4 TRIO — Fri 16 Oct · 16:00 to 16:20 · Room B

4.5 Deconstructing a transformer block — AI · 20 MIN

Dr. Collins Oduor Ondiek
Aicha Zindamoyen
Joy Dorcas Bisieri

Group 5 GROUP OF 4 — Thu 15 Oct · 16:20 to 17:00 · Room A

0.9 Responsible, not responsible, uncertain — RI · 40 MIN

Dr. Edward Ombui
Dr. Fidelis Kilonzo
Hillary Kavagi
Stanley K. Kimaita

Group 6 GROUP OF 4 — Thu 15 Oct · 16:20 to 17:00 · Room B

0.10 Context, perspective and LLMs — RI · 40 MIN

Dr. Joseph Onderi Orero
Dr. Eric Araka
Hilda Manegene
Moses Mataara

Group 7 GROUP OF 4 — Fri 16 Oct · 16:20 to 16:35 · Room A

1.8 Can versus should with LLMs — RI · 15 MIN

Dr. Kennedy Kilel
Joel Mutevu
Daniel Simiyu
Mary Wairimu Ndirangu

Group 8 GROUP OF 4 — Fri 16 Oct · 16:20 to 17:05 · Room B

1.10 The Ubuntu lens: individual and collective values — RI · 45 MIN

Dr. Rogers Ochenge
Victor Kemboi
Paul Kweyu
Susan Gitau

Questions? 🤔